

Post-Quantum Handshake (PQ-derived PSK, hybrid never PQ-only)

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Master technical specification — Volume 5 (Security & Privacy), nano-detail spec. Deepens invariant **S10** and §9 of [04-sec] (04-security-privacy-pki.md): the **post-quantum handshake**. This document owns the *byte-level and operational contract* of HelixVPN's harvest-now-decrypt-later (HNDL) defence — the PQ KEM exchange, the HKDF→WireGuard-PSK derivation, the **hybrid-never-PQ-only** rule, the capability negotiation, the rekey lifecycle, the KEM decision (ML-KEM-768 vs Rosenpass vs Classic-McEliece-hybrid), the wire framing, the rollout phasing, and the anti-bluff validation. This is a **SPEC** — it describes the implementation; it does not build the product.

The PQ exchange does **not** modify the WireGuard protocol: it fills WireGuard's *already-present, optional 32-byte preshared-key (PSK) slot* — the single injection point. The PSK mechanics, key schedule, and the PeerConfig.psk field live in the data plane (helix-wg); this document references them rather than redefining them.

Source evidence cited inline by id: [04-sec §9] (04-security-privacy-pki.md §9 + S10 + threat T8), [wg-core §4.4] (v02-data-plane/wireguard-core.md PQ-PSK seam), [wg-core §2.2/§4.2] (the PeerConfig.psk field + key-schedule mix-in), [research-pq] (v09-research/research-pki_pq_nat.md §2 post-quantum WireGuard), [research-wg §6] (v09-research/research-wireguard.md §6 the PSK injection point), [research-mullvad] (the PQ-PSK-over-WG prior art). Access date for all external facts: **2026-06-25**. Unproven claims are marked UNVERIFIED: per §11.4.6 — never fabricated.

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0. Position, ownership, and the one-sentence contract

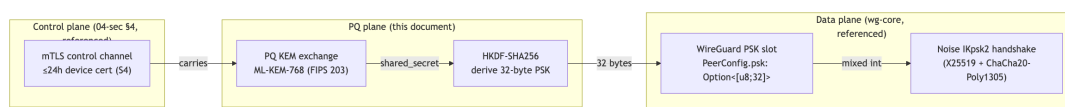
One-sentence contract. HelixVPN derives a 32-byte symmetric pre-shared key from a post-quantum KEM exchange and mixes it into WireGuard's existing optional-PSK slot, so a session is secure unless an attacker breaks **both** the classical X25519 ECDH **and** the PQ KEM — closing the HNDL threat without changing the WireGuard protocol [04-sec §9, wg-core §4.4, research-pq §2.1].

What this document owns:

- the PQ KEM exchange protocol that runs *over the authenticated control channel*;
- the HKDF derivation `shared_secret` $\rightarrow$ 32-byte WireGuard PSK;
- the hybrid combiner (classical + PQ, S10) and the never-PQ-only rule;
- the capability negotiation + classical-WG fallback;
- the per-rekey PSK rotation lifecycle;
- the KEM decision (ML-KEM-768 default; Rosenpass / Classic-McEliece-hybrid options);
- the `PqKem` pluggable Rust contract and the wire framing.

What this document does NOT own (referenced, not redefined):

- the WireGuard Noise IK handshake, the PSK key-schedule mix-in, and the `PeerConfig.psk: Option<Psk>` field — those are [wg-core §4.2/§4.4/§2.2]; this doc only *fills* the field;
- the control channel mTLS + device-cert lifecycle the PQ exchange rides on — [04-sec §4];
- where the PQ code physically lives in `helix-core` crates + FFI — Volume 5 client-core (the `helix-pki` crate's `pq` module); this doc defines the *protocol*, not the file tree.



1. The threat: harvest-now-decrypt-later (HNDL)

WireGuard's handshake is **Noise_IK** over **Curve25519** (X25519) ECDH [research-wg §1, §2]. X25519 is classically secure but **not** post-quantum secure: a sufficiently large quantum computer running Shor's algorithm recovers the ECDH private key from observed public values, decrypting any session whose key agreement used it.

The operative danger is **not** "quantum computers exist today" — they do not at the scale required (§12 residual risk). The danger is **harvest-now-decrypt-later**: a passive adversary (a censor, an ISP, a nation-state recorder) **records ciphertext today** and **decrypts it years later** once a cryptographically-relevant quantum computer matures [research-pq §2.1, 04-sec §9]. For a privacy VPN this is the worst class of threat because:

1. The victim cannot detect the recording — it is passive.
2. The damage is retroactive — every session recorded before the mitigation shipped is exposed; you cannot re-secure traffic an adversary already holds.
3. The window is long — HNDL captures accumulate for as long as the adversary chooses, and the decryption deadline is the adversary's, not the victim's.

This is exactly the threat invariant **S10** binds and threat **T8** enumerates in [04-sec §0.1, §10]: *passive recorder + future quantum*, mitigated by the hybrid PQ PSK, residual risk = "PQ KEM novelty → hybrid hedge; classical still protects."

Honest boundary (§11.4.6). PQ protection here defends *future* decryption of *today's* recorded traffic. It is **not** a claim that classical WireGuard is broken now — X25519 + ChaCha20-Poly1305 is sound against every classical adversary [research-wg §2]. The PQ layer is a forward hedge, deliberately conservative.

2. The mechanism: WireGuard's PSK slot is the only injection point

WireGuard already supports an **optional 32-byte symmetric pre-shared key**. The protocol page is explicit: *"When pre-shared key mode is not in use, the pre-shared key value used below is assumed to be an all-zero string of 32 bytes"* [research-wg §6] — i.e. the PSK slot **always exists**; it is simply all-zero when unused. In the Noise_IKpsk2 variant the PSK is mixed into the chaining key during response generation: `temp = HMAC(responder.chaining_key, preshared_key)` [research-wg §6].

Reconciled (§11.4.35, 2026-06-26): this key-schedule formula is cited to [research-wg §6] only; the prior dual-citation to `wireguard-core.md §4.2` is dropped because `wg-core §4.2` marks its key-schedule transcription **UNVERIFIED** — the authoritative source for the PSK mix-in is the WireGuard protocol research §6.

The PSK enters the **symmetric** key schedule (BLAKE2s/HMAC chaining), *downstream* of the X25519 ECDH. The cryptographic consequence is the whole reason this works:

Because the PSK is mixed into the symmetric chaining key, the tunnel stays confidential **even if the X25519 ECDH is later broken** — an attacker who recovers the ECDH secret still faces an unknown 32-byte symmetric secret folded into the key schedule. [research-wg §6]

So the injection point is **exactly one field** and the WireGuard protocol surface needs **zero change** — only the PSK field is filled with a value derived from a PQ KEM, instead of being left all-zero or a static manual key [research-pq §2.1, research-wg §6].

In HelixVPN's data plane this is the single seam `helix-wg` exposes:

```
// referenced from v02-data-plane/wireguard-core.md §2.2 – NOT
// redefined here.
pub type Psk = [u8; 32]; // optional pre-shared
// key (post-quantum seam, §4.4)

pub struct PeerConfig {
    // ... allowed_ips, endpoint, persistent_keepalive,
    // public_key ...
}
```

```

    pub psk: Option<Psk>, // §4.4 – THIS is where
                          the PQ-derived PSK lands
}

```

and the underlying boringtun `Tunn::new` takes the same slot directly [research-wg §10]:

```

// boringtun (referenced) – the PSK slot at the implementation
// floor.
pub fn new(
    static_private: StaticSecret,
    peer_static_public: PublicKey,
    preshared_key: Option<[u8; 32]>, // ← the PQ-derived PSK
                                   installs here
    persistent_keepalive: Option<u16>,
    index: u32,
    rate_limiter: Option<Arc<RateLimiter>>,
) -> Self

```

The UAPI configures it as the `preshared_key` config key (hex; all-zero to remove) [research-wg §6]. HelixVPN's only data-plane work is to *populate* `psk` from the PQ exchange; everything below `PeerConfig.psk` is unmodified, well-audited WireGuard.

3. The hybrid model – classical X25519 ⊕ PQ KEM, never PQ-only (S10)

Invariant S10: PQ is hybrid, never PQ-only. The session derives security from *both* a classical ECDH **and** a PQ KEM, combined so an attacker must break both [04-sec §0.1 S10, research-pq §2.1].

The two halves and why each is load-bearing:

Half	Primitive	Protects against	Failure mode it covers
Classical	X25519 ECDH (WireGuard's native handshake)	every <i>classical</i> adversary today	a flaw discovered in the (younger, less battle-tested) PQ KEM
Post-quantum	ML-KEM-768 KEM (FIPS 203), default	a <i>future quantum</i> HNDL adversary	quantum break of X25519 ECDH

The combiner is **defence-in-depth**: the X25519 handshake runs exactly as WireGuard always runs it (the classical half is *intrinsic* — it is the Noise IK handshake itself), and the PQ KEM's shared secret is folded into the **same** session via the PSK slot. An attacker must break **both** the classical ECDH **and** the PQ KEM to recover the session keys [04-sec §9.1, research-pq §2.1].

Why never PQ-only (the conservative choice, §11.4.6). PQ KEMs are *younger* than X25519 and have had less cryptanalytic exposure. If the project shipped a PQ-only handshake and the chosen PQ KEM later proved flawed, *every* session would be exposed. With the hybrid:

- if the PQ KEM has a flaw → classical WireGuard still protects you against today's attackers;
- if quantum breaks classical X25519 → the PQ PSK still protects you against HNDL.

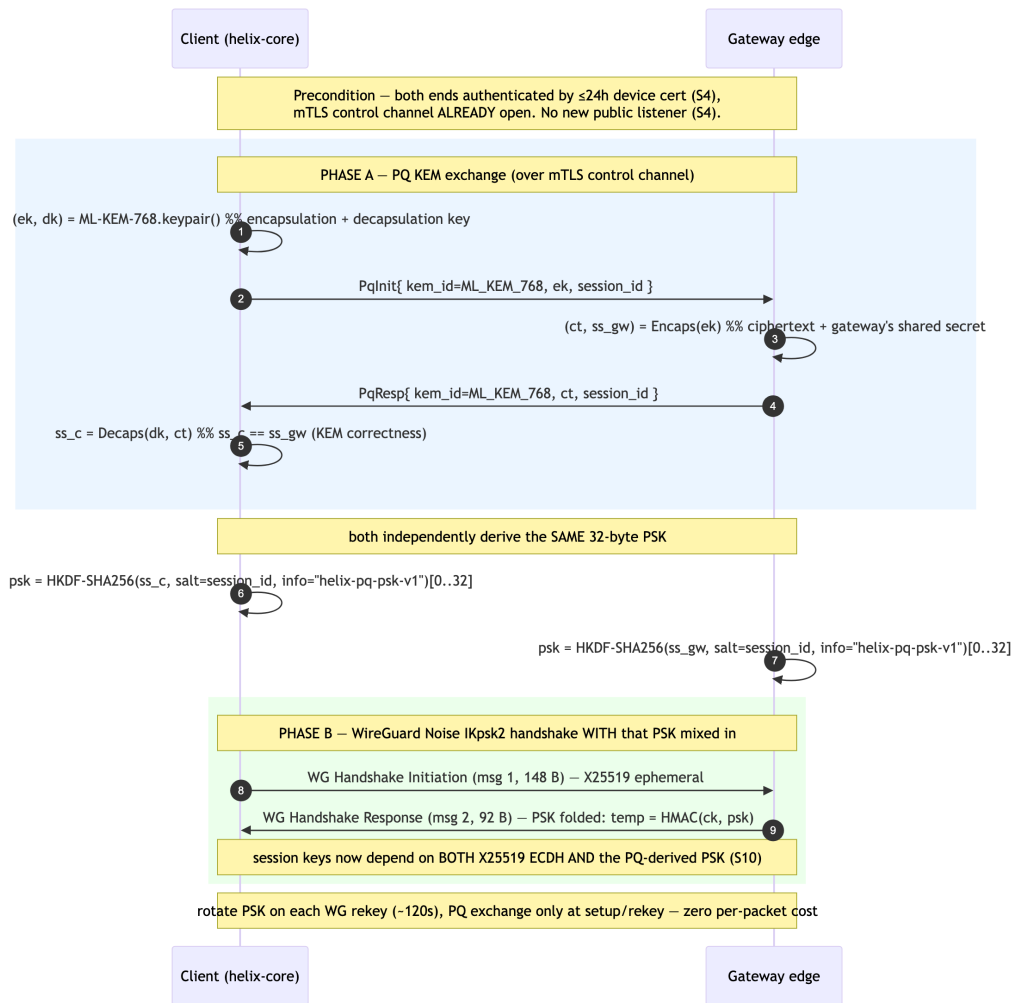
This is the standing industry recommendation — *"hybrid, not PQ-only... production deployments combine a classical KEM and a PQ KEM so the result is no weaker than either"* [research-pq §2.1] — and matches Mullvad's production posture (Classic McEliece + ML-KEM, hybrid by default) [research-pq §2.2, research-mullvad].

Combiner construction (the symmetric fold is the combiner).

HelixVPN does *not* build a bespoke KEM combiner construction. The classical half is the X25519 ECDH already inside the Noise IK handshake; the PQ half is the KEM shared secret installed as the WireGuard PSK. The WireGuard key schedule itself is the combiner: the final session keys depend on the X25519 chaining **and** the PSK mix-in temp = HMAC(chaining_key, psk) [research-wg §6]. Breaking one input does not yield the session keys. **UNVERIFIED**: the formal IND-CCA composition argument for "Noise IKpsk2 with a KEM-derived PSK is a sound hybrid" — this rests on the WireGuard whitepaper's Noise_IKpsk2 security argument, which a human must read directly [research-wg §"Honest gaps", wg-paper UNREACHABLE]; flagged, not claimed.

4. The hybrid handshake (Mermaid + step-by-step)

The PQ KEM exchange rides the **authenticated mTLS control channel** (S4) — there is **no new public listener** and no unauthenticated PQ endpoint to attack [04-sec §9.3, research-pq §1.2]. Both ends already hold a valid ≤24 h device cert (S4); the PQ exchange is one more RPC over that channel, before/alongside the WireGuard data handshake.



4.1 Step-by-step (the contract)

- Both ends authenticate first.** The mTLS control channel is already open with valid ≤ 24 h device certs (S4) [04-sec §4]. PQ negotiation is gated on this — an unauthenticated party never reaches the PQ exchange.
- Capability check (§7).** Both ends advertise `pq_capable=true` + the supported `kem_id` set. If either end lacks PQ support, the session falls back to classical WireGuard (still secure today) and skips Phase A entirely [04-sec §9.3, research-pq §2.2 mobile-experimental].
- PQ KEM exchange (Phase A).** The client generates an ML-KEM-768 keypair (ek, dk) , sends the **encapsulation key** `ek` to the gateway. The gateway runs $\text{Encaps}(ek) \rightarrow (ct, ss_{gw})$, returns the **ciphertext** `ct`. The client runs $\text{Decaps}(dk, ct) \rightarrow ss_c$. By KEM correctness, $ss_c == ss_{gw}$ [research-pq §2.4]. The client's *decapsulation* key `dk` never leaves the device.

4. **PSK derivation.** Both ends independently compute $\text{psk} = \text{HKDF-SHA256}(\text{shared_secret}, \text{salt}=\text{session_id}, \text{info}=\text{b"helix-pq-psk-v1"})$ truncated to 32 bytes [04-sec §9.3 code, RFC 5869]. The `session_id` salt domain-separates each session so the same KEM material never yields the same PSK twice.
5. **WireGuard handshake with the PSK (Phase B).** The standard Noise IKpsk2 handshake runs; the 32-byte PSK is mixed into the chaining key $\text{temp} = \text{HMAC}(\text{ck}, \text{psk})$ during response generation [research-wg §6]. The resulting session keys depend on **both** the X25519 ECDH **and** the PQ-derived PSK (§10).
6. **Rotation.** On every WireGuard rekey ($\text{REKEY_AFTER_TIME} \approx 120 \text{ s}$ [research-wg §9]) the PQ exchange re-runs and a fresh PSK is installed (§8). The PSK is *ephemeral* — never persisted (§4.5 of [04-sec], the PQ plane "per-session, never persisted").

4.2 What the adversary sees

A passive recorder on the data path sees only **opaque WireGuard datagrams** carried by the obfuscation transport (doc 01) — the PQ exchange happened on the *encrypted, authenticated mTLS control channel*, so the recorder cannot even observe the KEM ciphertext, let alone the PSK [04-sec §9.3]. Even granting the adversary the KEM ciphertext, recovering the session requires breaking ML-KEM-768 **and** X25519 (§10).

5. KEM choice — decision with recommendation

Per §11.4.66 (decision discipline) the KEM is surfaced as a **decision with options**, never a silent pick. The PqKem trait (§10) makes the choice pluggable so the decision is reversible.

Option	KEM(s)	Pro	Con	Recommendation
Primary	ML-KEM-768 (FIPS 203, standardized Kyber)	NIST-standardized (FIPS 203, published 2024-08-13); good size/speed balance; broad library support (ml-	younger than X25519; lattice assumption is newer cryptanalysis	DEFAULT. Pluggable in helix-core's pq module [04-sec §9.2, research-pq §2.4].

Option	KEM(s)	Pro	Con	Recommendation
Conservative hybrid	ML-KEM-768 + Classic McEliece	kem Rust crate, BoringSSL); NSA CNSA 2.0 mandates ML-KEM for NSS key-encapsulation	Mullvad's production posture combines McEliece + ML-KEM for exactly this hedge; code-based (McEliece) is a <i>very different</i> hardness assumption from lattice (ML-KEM) — a break of one does not imply a break of the other	Phase-2 opt-in for high-assurance tenants; runs only at setup/rekey so the size cost is amortised, not per-packet [04-sec §9.2, research-pq §2.2].
			Classic McEliece public/encapsulation keys are large (hundreds of KB) → larger control-channel exchange	
External protocol	Rosenpass (audited PQ key-exchange daemon feeding WG a PSK)	independent symbolic analysis (ProVerif); clean separation of concerns; reference design Mullvad embeds in its agent (since v0.25.4) auto-rotating PSKs	extra daemon + integration surface; tagged release (v0.2.2, 2024-06-05) still uses Kyber-512 + Classic McEliece , <i>not</i> yet ML-KEM in the tagged build; full cryptographic proof "in	Evaluate during the Phase-2 PQ spike — prefer Rosenpass if an <i>audited external protocol</i> beats an in-house KEM exchange; verify upstream ML-KEM status before relying on it [04-sec §9.2, research-pq §2.3].

Option	KEM(s)	Pro	Con	Recommendation
			progress", not complete	

5.1 Decision rationale

- **ML-KEM-768 as the default** because it is the *standardized* primitive (FIPS 203), has the cleanest Rust library story (`ml-kem` crate), and gives the best size/speed for a per-rekey exchange. NSA CNSA 2.0 requiring ML-KEM for national-security systems means the ecosystem and tooling will converge on it [research-pq §2.4].
- **Why not 1024.** Mullvad uses ML-KEM-**1024** in its current production tunnels [research-wg §6 cross-ref]. HelixVPN's MVP/Phase-2 default is **768** for the size/speed balance, with the parameter set being a `kem_id` field (§9) so a tenant can negotiate 1024 if it wants the larger margin. UNVERIFIED: whether 768 vs 1024 is the right default for HelixVPN's threat model at GA — this is a Phase-2 spike decision; 768 is the documented starting recommendation, not a final pin (§11.4.6).
- **Rosenpass vs in-house.** The genuine question the Phase-2 spike answers: is it better to embed an *audited external* PQ protocol (Rosenpass, the path Mullvad took) or to drive ML-KEM directly via the PqKem trait? Rosenpass brings independent analysis and a maintained reference, at the cost of an extra daemon and a tagged release that is **not yet on ML-KEM**. The decision is deferred to the spike with captured evidence, not pre-judged here [04-sec §9.2, research-pq §2.3].

Honest gap (§11.4.6 / §11.4.99). Rosenpass's verification status is a **symbolic analysis (ProVerif)**, **NOT a full cryptographic proof** — the proof is "in progress" per its README [research-pq §2.3]. The tagged Rosenpass release (v0.2.2, 2024-06-05) still uses Kyber-512, not ML-KEM. Both facts **MUST** be re-verified against upstream at implementation time per §11.4.99 (latest-source) before relying on Rosenpass-specific ML-KEM behaviour.

6. Key sizes, performance, and the steady-state-zero-cost claim

The defining performance property: **the PQ exchange runs at session setup and rekey only — never per packet** — so steady-state overhead is **zero** [04-sec §9.3, research-pq §2.4].

6.1 Approximate key/ciphertext sizes (orientation, verify at impl)

KEM	Encapsulation key	Ciphertext	Shared secret	Note
ML-KEM-768	UNVERIFIED: $\approx$ 1184 B	UNVERIFIED: $\approx$ 1088 B	32 B	FIPS 203 Module-Lattice KEM; verify exact sizes against the <code>ml-kem</code> crate / FIPS 203 at impl time
ML-KEM-1024	UNVERIFIED: $\approx$ 1568 B	UNVERIFIED: $\approx$ 1568 B	32 B	larger margin; negotiable via <code>kem_id</code>
Classic McEliece	hundreds of KB (large public key)	small	32 B	the size cost is at <i>setup/rekey only</i> , amortised [research-pq §2.2]
Rosenpass (Kyber-512 + McEliece)	dominated by McEliece public key	—	32 B $\rightarrow$ WG PSK	external daemon; refreshes PSK every 2 min [research-pq §2.3]

UNVERIFIED: byte sizes. The exact ML-KEM-768/1024 key and ciphertext byte sizes are standardized in FIPS 203 but are **not** re-confirmed from a fetched primary source in this writing context (the figures above are orientation only) — they **MUST** be pinned against the chosen `ml-kem` crate version and FIPS 203 at implementation time per §11.4.6/§11.4.99.

6.2 Cost model

- **Per-packet cost: zero.** The PQ exchange does not touch the data path. Once the PSK is installed, every transport-data packet is plain ChaCha20-Poly1305 with the PSK-influenced session key — identical cost to classical WireGuard [04-sec §9.3].

- **Per-rekey cost: KB-scale exchange + one Encaps/Decaps.** ML-KEM Encaps/Decaps are fast (microsecond-class on commodity hardware) and the exchange is KB-scale; with REKEY_AFTER_TIME ≈ 120 s [research-wg §9] this is one small exchange every ~ 2 minutes per peer.
 - **Handshake-time budget.** The PQ-PSK exchange adds roughly $\sim 1\text{--}2$ s **at handshake**, with **steady-state unchanged** [wg-core §"perf table" / research-mullvad §4]. HelixVPN's handshake budget already allots for this (handshake < 1 s plain, < 2 s MASQUE) [wg-core §15].
 - **Classic McEliece caveat.** McEliece's hundreds-of-KB public key makes its *setup/rekey* exchange materially larger; this is why it is the Phase-2 high-assurance opt-in, not the default, and why it runs only at setup/rekey where the size is amortised [research-pq §2.2].
-

7. Negotiation, capability flags, and fallback

PQ is **capability-negotiated** with a graceful classical fallback — a peer that does not support PQ falls back to classical WireGuard, which is still secure today [04-sec §9.3, research-pq §2.2].

7.1 The toggle and the default

- A "**Quantum-resistant**" **toggle** is exposed to the user, **on by default for new tunnels where both ends support it** [04-sec §9.3]. This mirrors Mullvad's posture: quantum-resistant tunnels became the **default on desktop as of 2025-01-09** [research-pq §2.2].
- **Mobile honesty (§11.4.6).** Mullvad shipped PQ for iOS, but its embedded Rosenpass-server PSK auto-rotation path is still marked **experimental and not yet supported on mobile** in the agent [research-pq §2.2]. HelixVPN treats mobile PQ as Phase-2 with the same honesty: the toggle exists, the on-by-default policy is platform-gated, and a platform where the PQ path is not yet proven ships PQ as opt-in, not silent default — never a faked "PQ on" that is actually classical (§11.4.6 no-guessing).

7.2 Negotiation outcomes (closed set)

Client PQ	Gateway PQ	Outcome	PSK	Security posture
supports + toggle on	supports	PQ hybrid	PQ-derived (this doc)	$X25519 \oplus$ PQ KEM (S10)

Client PQ	Gateway PQ	Outcome	PSK	Security posture
supports + toggle off	supports	classical WG	all-zero (none)	X25519 only — secure today, no HNDL hedge
supports	does not support	classical WG (fallback)	all-zero	X25519 only — graceful degrade, never a faked PQ PASS
does not support	supports	classical WG (fallback)	all-zero	X25519 only

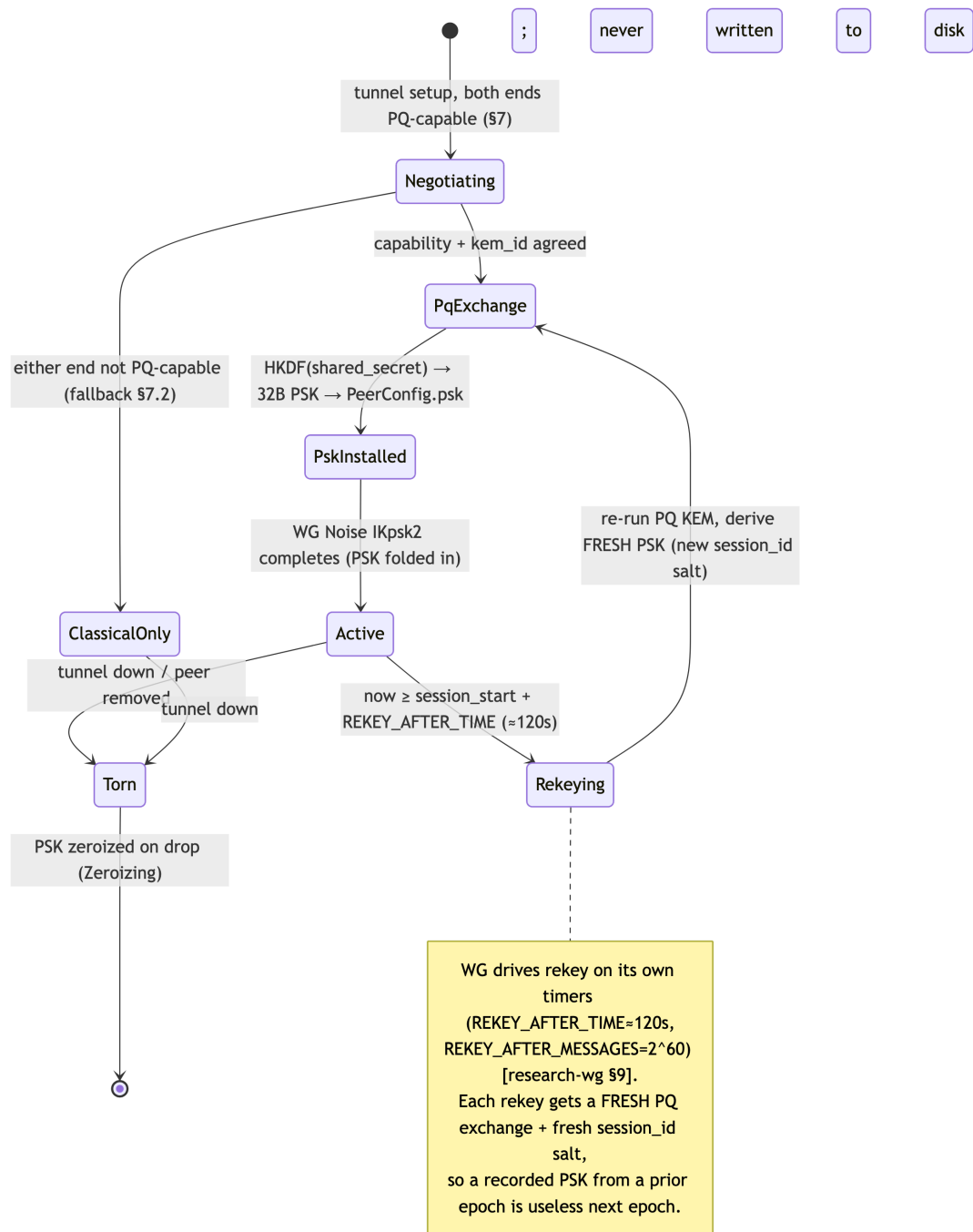
The negotiation result is **observable**: the runtime signature for "PQ is actually on" is *the WireGuard PSK is non-zero AND an ML-KEM exchange was observed* (§13). A toggle that reads "on" while the PSK is all-zero is a §11.4 PASS-bluff and a §13 test catches it.

7.3 kem_id negotiation

Both ends advertise their supported kem_id set (ML-KEM-768, optionally ML-KEM-1024, optionally McEliece-hybrid, optionally Rosenpass). They select the **strongest mutually supported** KEM. An unknown kem_id is rejected (closed-set, §9) — never silently downgraded to a weaker one without recording the negotiation outcome [§11.4.6].

8. Rekey & PSK rotation lifecycle

The PQ-derived PSK is **ephemeral and rotated on every WireGuard rekey** — it is never persisted [04-sec §4.5 "PQ PSK rotated on every WG rekey interval; ephemeral, never persisted"].



8.1 Rotation contract

- **Trigger.** WireGuard's own rekey timers drive rotation:
 $\text{REKEY_AFTER_TIME} \approx 120 \text{ s}$ or $\text{REKEY_AFTER_MESSAGES} = 2^{60}$ [research-wg §9]. The PQ layer does *not* invent its own rotation cadence — it reuses WireGuard's, so the timings are never guessed (§11.4.6).
- **Fresh material each epoch.** Every rekey runs a **new** PQ KEM exchange with a **new** session_id salt, yielding a fresh PSK. A PSK recorded during epoch N (even if somehow extracted) is useless in epoch $N+1$ — forward secrecy at the PSK layer.

- **Ephemerality (S11-class hygiene).** The PSK lives only in memory, wrapped in Zeroizing so it is wiped on drop; it is **never** written to disk, never logged, never serialized across the FFI boundary as raw bytes [04-sec §3.3 key-handling discipline, §4.5]. WireGuard's own session zeroization (`REJECT_AFTER_TIME × 3`) wipes the derived session keys [research-wg §9].
- **Rosenpass cadence cross-ref.** If Rosenpass is chosen (§5), it refreshes the PSK **every 2 minutes** on its own [research-pq §2.3] — which aligns with WireGuard's ~120 s rekey, so the two cadences compose naturally. **UNVERIFIED:** exact interaction of Rosenpass's 2-min refresh with WireGuard's rekey timing under HelixVPN's specific wiring — a Phase-2 integration detail.

9. Wire framing of the PQ exchange (on the mTLS control channel)

The PQ exchange is **two messages** over the authenticated mTLS control channel — PqInit (client → gateway, carries the encapsulation key) and PqResp (gateway → client, carries the KEM ciphertext). It is an extension of the Coordinator RPC surface [04-sec §3.2 extends doc 03], so it inherits mTLS authentication and needs no new listener.

```
// helix-proto/coordinator.proto (PQ additions – extends the
Coordinator service)
// All PQ RPCs ride the SAME mTLS control channel as Enroll/
WatchNetworkMap (S4). No new listener.

enum KemId {
    KEM_UNSPECIFIED = 0; // reject (closed-set; never silently
downgrade)
    KEM_ML_KEM_768   = 1; // DEFAULT (FIPS 203)
    KEM_ML_KEM_1024  = 2; // larger margin, negotiable
    KEM_MCELIECE_HYB = 3; // Phase-2 ML-KEM-768 + Classic
McEliece (high-assurance opt-in)
    KEM_ROSENPASS     = 4; // Phase-2 external Rosenpass protocol
(evaluated in the spike)
}

message PqCapabilities { // exchanged during
negotiation (§7)
    bool pq_capable = 1; // false ⇒ classical
fallback, never a faked PQ PASS
    repeated KemId supported = 2; // strongest mutually-
supported wins
    bool toggle_on = 3; // user "Quantum-
resistant" toggle state (§7.1)
}
```

```

message PqInit {                                     // client → gateway
    (PHASE A step 1)
    KemId    kem_id    = 1;                          // selected per §7.3;
    KEM_UNSPECIFIED ⇒ PermissionDenied
    bytes    encap_key  = 2;                          // ML-KEM-768
    encapsulation key (ek); size per §6.1
    bytes    session_id = 3;                          // 16-byte random; the
    HKDF salt (domain-separates epochs)
}

message PqResp {                                     // gateway → client
    (PHASE A step 2)
    KemId    kem_id    = 1;                          // MUST echo
    PqInit.kem_id; mismatch ⇒ abort
    bytes    ciphertext = 2;                          // ML-KEM ciphertext
    (ct); client Decaps(dk, ct) → ss
    bytes    session_id = 3;                          // MUST echo
    PqInit.session_id
}
// NOTE: the decapsulation key (dk) and the derived 32-byte PSK
// NEVER appear on the wire (S2-class).
// Only ek (public) and ct (public) cross the channel; ss and psk
// are computed independently each side.

```

9.1 Framing invariants

1. **Public material only on the wire.** Only ek (encapsulation key, public) and ct (ciphertext, public) cross the channel. The decapsulation key dk and the derived psk are computed independently on each side and never transmitted (S2-class key hygiene).
 2. **session_id binds the epoch.** The 16-byte random session_id is the HKDF salt; echoing it in PqResp binds request↔response and domain-separates each rekey epoch.
 3. **Closed kem_id set.** KEM_UNSPECIFIED (0) is rejected; an unknown kem_id aborts the exchange — never a silent downgrade to a weaker KEM (§11.4.6).
 4. **UNVERIFIED: exact PQ-PSK message sequence.** The precise KEM-exchange framing is a HelixVPN design choice — Mullvad's equivalent is *deliberately high-level in public docs* and is "a Mullvad-specific protocol on top of WireGuard ... NOT part of the WireGuard protocol itself" [research-wg §6, research-pq §2.2]. The two-message PqInit/PqResp shape above is HelixVPN's spec design, not a copy of a published wire format; it MUST be reviewed against the chosen KEM library's API and (if Rosenpass) Rosenpass's own protocol at impl time.
-

10. Pluggable KEM contract (Rust)

The KEM is behind a trait so the §5 decision is reversible and the §13 self-validation can swap a golden-good / golden-bad implementation [04-sec §9.3].

```
// helix-core/crates/helix-pki/src/pq.rs (pluggable KEM contract
// - SPEC, not full impl)

/// A post-quantum KEM. The default impl is ML-KEM-768 (FIPS 203).
/// Pluggable so the §5 decision
/// (ML-KEM vs McEliece-hybrid vs Rosenpass) is a swap, not a
/// rewrite, and so the analyzer
/// self-validation (§13) can inject a golden-good and a golden-
/// bad KEM.
pub trait PqKem {
    fn kem_id(&self) -> KemId;
    /// Generate an (encapsulation key, decapsulation key) pair.
    /// dk NEVER leaves the device.
    fn keypair(&self) -> (EncapKey, DecapKey);
    /// Gateway side: encapsulate to the client's ek, producing
    /// (ciphertext, shared_secret).
    fn encapsulate(&self, ek: &EncapKey) -> (Ciphertext,
        SharedSecret);
    /// Client side: decapsulate the ciphertext with dk,
    /// recovering the SAME shared_secret.
    fn decapsulate(&self, dk: &DecapKey, ct: &Ciphertext) ->
        SharedSecret;
}

/// PSK = HKDF-SHA256(shared_secret, salt=session_id,
/// info=b"helix-pq-psk-v1") -> 32 bytes,
/// installed into the WireGuard Noise IKpsk2 preshared-key slot
/// (PeerConfig.psk, wg-core §2.2/§4.4).
/// Rotated every WG rekey (§8). The returned WgPsk is Zeroizing -
/// wiped on drop, never persisted.
pub fn derive_wg_psk(ss: &SharedSecret, session_id: &[u8]) ->
    WgPsk {
    // HKDF-Extract(salt=session_id, ikm=ss) -> prk; HKDF-
    // Expand(prk, "helix-pq-psk-v1", 32) -> psk
    // (RFC 5869). The 32-byte output is exactly the WireGuard PSK
    // width.
    /* HKDF-SHA256 */ unimplemented!("contract - impl batched per
    §11.4.9")
}

/// The hybrid combiner is NOT a bespoke construction: the
/// classical X25519 half is intrinsic to
/// the Noise IK handshake; this PSK is the PQ half folded into
/// the SAME session key schedule
/// (wg-core §4.2 `temp = HMAC(ck, psk)`). Breaking one input
/// alone does not yield the session keys (S10).
```

10.1 Boundary discipline

- EncapKey and Ciphertext are public, serialized onto the wire (§9). DecapKey and SharedSecret and WgPsk are secret: Zeroizing-wrapped, never serialized, never logged, never crossing the FFI boundary as raw bytes [04-sec §3.3].
- The trait is the seam for the §13 self-validation: a **golden-bad** PqKem whose decapsulate returns a value $\neq$ the gateway's encapsulate shared secret **MUST** make the handshake **fail** (PSK mismatch $\Rightarrow$ WG handshake fails) — proving the test catches a broken KEM, not just agreeing with a working one (§11.4.107(10)).

11. Rollout phasing (Phase 2+) and standardized-vs-experimental honesty

PQ is **Phase 2+**, not MVP — the MVP ships classical WireGuard (secure today) and reserves the PSK seam so PQ is purely additive [04-sec §9, §12 Phase-2 plan, wg-core §4.4 "additive, Phase 2"].

Phase	Scope	Status honesty (§11.4.6)
MVP (Phase 1)	Classical WireGuard only; PSK slot present but all-zero; PqCapabilities reserved in the proto	STANDARDIZED + shipping — X25519 Noise IK is mature, well-audited [research-wg §2]
Phase 2	PqKem ML-KEM-768 over mTLS; HKDF→WG PSK; capability negotiate; rotate on rekey; "Quantum-resistant" toggle on-by-default where both ends support; spike: ML-KEM vs Rosenpass vs McEliece-hybrid	ML-KEM-768 is STANDARDIZED (FIPS 203, 2024-08-13); the <i>HelixVPN wire framing</i> (§9) is design work , not a standard; Rosenpass is EXPERIMENTAL (symbolic-only analysis) [research-pq §2.3]
Phase 2 (high-assurance opt-in)	ML-KEM-768 + Classic McEliece hybrid for tenants wanting the code-based hedge	McEliece is a NIST-approved primitive; the <i>combination</i> matches Mullvad production [research-pq §2.2]
Phase 3	Reproducible builds + cosign-signed images + SBOM + third-party audit of the PQ core (so users can	the PQ claim is only as trustworthy as the auditable build

Phase	Scope	Status honesty (§11.4.6)
	verify the PQ claim) [04-sec §12.3 Phase 3]	

11.1 Standardized vs experimental — the honest split

- **Standardized (reliable now):** ML-KEM (FIPS 203, published 2024-08-13); the WireGuard PSK mechanism (the PSK slot is part of the WireGuard protocol) [research-pq §2.4, research-wg §6].
- **Design work (HelixVPN-specific, must be reviewed):** the §9 PqInit/PqResp wire framing; the capability-negotiation flow (§7); the HKDF info-string "helix-pq-psk-v1" and the PSK derivation parameters. These are HelixVPN's protocol design, not a published standard.
- **Experimental (do not over-claim):** Rosenpass — symbolic analysis (ProVerif) only, full cryptographic proof "in progress", tagged release still on Kyber-512 [research-pq §2.3]; mobile PQ auto-rotation (Mullvad marks it experimental-not-yet-supported on mobile) [research-pq §2.2]. Any HelixVPN doc/UI that claims these as production-grade is a §11.4.99 stale/over-claim violation — re-verify against latest upstream before relying on them.

12. Threat-model placement & residual risk

PQ closes threat **T8** in [04-sec §10]:

#	Threat (STRIDE)	Vector	Mitigation (this doc)	Residual risk (§11.4.6)
T8	Information disclosure — harvest-now-decrypt-later	Passive recorder + future quantum computer	Hybrid PQ PSK (§10, §3–§9): X25519 ⊕ ML-KEM-768, PSK rotated per rekey	PQ KEM novelty → the <i>hybrid hedge</i> covers it (classical still protects if the PQ KEM is flawed); a break of <i>both</i> X25519 <i>and</i> ML-KEM is required to recover a session

Residual risks stated honestly:

1. **PQ KEM novelty.** ML-KEM is younger than X25519; a future cryptanalytic break is possible. The hybrid design (§3) is *exactly* the hedge: a PQ-KEM break leaves classical WireGuard intact for today's adversaries.
 2. **HelixVPN wire-framing risk.** The §9 PqInit/PqResp framing is design work, not a standard — a protocol flaw in *the framing* (not the KEM) is possible and is why §13 + a third-party audit (Phase 3) are required before over-claiming.
 3. **Mobile/Rosenpass maturity.** Where the PQ path is experimental (mobile, Rosenpass), PQ is opt-in, not silent default — never a faked "PQ on."
 4. **Formal-composition gap.** The IND-CCA composition argument for Noise IKpsk2 + KEM-derived PSK rests on the WireGuard whitepaper's security argument, which was **UNREACHABLE** via the fetch tool and must be read directly by a human [research-wg §"Honest gaps"]. Flagged, not claimed.
 5. **Not a claim classical WG is broken now.** PQ is a *forward* hedge; X25519 + ChaCha20-Poly1305 is sound against every classical adversary today [research-wg §2].
-

13. Anti-bluff validation (how PQ is proven, not asserted)

Per S10's verification row [04-sec §13] and §11.4.107/§11.4.69, "PQ is on" is proven by **captured evidence**, never by a toggle reading "on."

Invariant	Captured-evidence proof (§11.4.5/§11.4.69/§11.4.107)
PQ hybrid active	Handshake with PQ on $\Rightarrow$ the WireGuard PSK is non-zero AND an ML-KEM exchange is observed on the mTLS channel (captured PqInit/PqResp + non-zero PeerConfig.psk). A toggle "on" with an all-zero PSK is a §11.4 PASS-bluff and FAILs.
Classical fallback works	PQ off (or peer not PQ-capable) $\Rightarrow$ classical WireGuard still connects (captured handshake), proving the fallback path is real and never a faked PQ PASS [04-sec §13].
Hybrid, not PQ-only	With PQ on, the X25519 handshake is <i>still present</i> (captured msg-1/msg-2) AND the PSK is non-zero — both

Invariant	Captured-evidence proof (§11.4.5/§11.4.69/§11.4.107)
	halves observed, proving the session depends on both (S10).
Rotation	Across a forced rekey, the captured PSK at epoch $N+1 \neq$ epoch N (fresh PQ exchange), proving rotation is real, not a static reused PSK.
KEM correctness	$\text{Decaps}(\text{dk}, \text{ct}) == \text{Encaps}(\text{ek}).\text{shared_secret}$ over N runs (§11.4.50 deterministic-consistency).
Self-validated analyzer	A golden-bad PqKem whose decapsulate returns the wrong shared secret MUST make the WG handshake FAIL (PSK mismatch) — proving the PQ test catches a broken KEM, not merely agreeing with a working one (§11.4.107(10), paired §1.1 mutation).

The PQ validation runs as `helix_qa + challenges suites` (§11.4.27) — a Challenge that scores PASS on a non-functional PQ control is the same class of defect as a unit-test PASS-bluff [04-sec §13]. PQ is *not* in the highest-risk-first set (kill-switch + revocation are, per §11.4.132) because it is Phase-2 additive, but its self-validation gate (golden-bad KEM $\Rightarrow$ handshake fails) is mandatory before any "PQ on" claim ships.

14. Phase $\rightarrow$ task $\rightarrow$ subtask plan (PQ workstream)

Each leaf becomes a §11.4.93 workable item. Anti-bluff: every closure carries captured evidence (§11.4.5/§11.4.69/§11.4.107) and a paired §1.1 mutation. Mirrors [04-sec §12 Phase-2 2.S1].

Phase 1 (MVP) — reserve the seam, ship classical

- **PQ-1.a** — `PeerConfig.psk`: `Option<Psk>` present in `helix-wg`; all-zero in MVP (the seam exists, unfilled) [wg-core §2.2/§4.4].
- **PQ-1.b** — `PqCapabilities/PqInit/PqResp/KemId` reserved in `coordinator.proto` so the wire format need not change later (§9; additive per §11.4.6 no-silent-reshape).

Phase 2 — PQ handshake

- **PQ-2.a** — PqKem trait + **ML-KEM-768** impl (`ml-kem` crate, pinned + re-verified per §11.4.99); `keypair/encapsulate/decapsulate` (§10).
- **PQ-2.b** — `derive_wg_psk` HKDF-SHA256 (`salt=session_id`, `info="helix-pq-psk-v1"`) $\rightarrow$ 32 B, Zeroizing (§4, §8).

- **PQ-2.c** — PqInit/PqResp exchange over the mTLS control channel; no new listener (§9).
- **PQ-2.d** — capability negotiation + classical fallback; "Quantum-resistant" toggle on-by-default where both support (§7); mobile gated as opt-in (§7.1 honesty).
- **PQ-2.e** — rotate PSK on every WG rekey; ephemeral, never persisted (§8).
- **PQ-2.f** — **spike**: ML-KEM-768 vs Rosenpass vs McEliece-hybrid decision (§5), captured evidence, deep-research per §11.4.150 before the structural pick.
- **PQ-2.g** — anti-bluff suite: non-zero-PSK + observed-ML-KEM-exchange evidence; golden-bad KEM $\Rightarrow$ handshake FAILs (§13); rotation evidence; N-iter KEM-correctness (§11.4.50).

Phase 2 (high-assurance opt-in)

- **PQ-2.h** — ML-KEM-768 + Classic McEliece hybrid PqKem impl for high-assurance tenants (§5); size-amortised at setup/rekey only (§6).

Phase 3 — auditable trust

- **PQ-3.a** — reproducible builds + cosign-signed images + SBOM so users verify the PQ binary matches source [04-sec §12.3].
- **PQ-3.b** — third-party cryptographic audit of the PQ core (KEM integration, HKDF derivation, wire framing, hybrid composition) [04-sec §12.3 3.S2].

Sources verified

- `final/04-security-privacy-pki.md` ([04-sec]) — §0.1 invariant **S10** (PQ hybrid, never PQ-only) + **S11** (PQ plane, ephemeral, never persisted); §9 (post-quantum handshake: PQ-derived PSK §9.1, hybrid handshake Mermaid §9.1, KEM choice table §9.2, where-it-runs + capability-negotiate + on-by-default + steady-state-zero-cost §9.3, PqKem/derive_wg_psk §9.3); §10 threat **T8** (HNDL) + residual risk; §13 PQ verification row; §12 Phase-2 task 2.S1 + Phase-3 reproducible builds/audit; §4.1 PQ plane key hierarchy; §4.5 PSK rotated on rekey, ephemeral. — local spec.
- `v02-data-plane/wireguard-core.md` ([wg-core]) — §2.2 (PeerConfig.psk: Option<Psk>, Psk = [u8;32]), §4.2 (key-schedule PSK mix-in temp = HMAC(ck, psk)), §4.4 (post-quantum PSK seam, additive Phase 2, no transport change, install negotiated shared secret as WG PSK), §15 (handshake budget), perf table (PQ-PSK ~1–2 s at handshake, steady-state unchanged). — local spec.

- `v09-research/research-pki_pq_nat.md` ([research-pq]) — §2.1 (PSK-injection approach, hybrid-not-PQ-only), §2.2 (Mullvad production: Classic McEliece + ML-KEM, default-on-desktop 2025-01-09, mobile experimental, wgephemeral peer reference, Kyber→ML-KEM migration), §2.3 (Rosenpass: Rust, PSK refresh every 2 min, Classic McEliece + Kyber-512, ProVerif symbolic analysis not full proof, v0.2.2 2024-06-05 still Kyber-512), §2.4 (FIPS 203 ML-KEM published 2024-08-13, NSA CNSA 2.0, target hybrid X25519 + ML-KEM-768, PSK feed not a WG fork). — cited research, access date 2026-06-25.
- `v09-research/research-wireguard.md` ([research-wg]) — §1/§2 (Noise IK, X25519, fixed crypto set), §6 (the PSK = post-quantum injection point: optional 32-byte PSK, all-zero when unused, `temp = HMAC(responder.chaining_key, preshared_key)`, enters symmetric schedule so secure even if X25519 broken; Mullvad ML-KEM-1024/McEliece-over-WG prior art; `PeerConfig.psk` surface; UAPI `preshared_key`; UNVERIFIED: exact PQ-PSK message sequence), §9 (rekey timers `REKEY_AFTER_TIME=120s`, `REKEY_AFTER_MESSAGES=2^60`, session zeroization), §10 (boringtun Tunn: :new preshared_key: `Option<[u8;32]>`), §"Honest gaps" (whitepaper PDF UNREACHABLE). — cited research, access date 2026-06-25.
- [research-mullvad] (via research-wg §6 / wg-core perf table cross-ref) — PQ-PSK-over-WG prior art; PQ-PSK ~1–2 s at handshake, steady-state unchanged.

Constitution bindings applied: §11.4.44 (revision header), §11.4.6 (no-guessing — every unproven size/framing/composition claim marked UNVERIFIED:; KEM choice + 768-vs-1024 + Rosenpass surfaced as decisions, never silent picks; rekey cadence reused from WireGuard, never invented), §11.4.66 (decision discipline — §5 KEM table with options + recommendation), §11.4.99 (latest-source — Rosenpass/ML-KEM status flagged for re-verification at impl), §11.4.107(10) (self-validated analyzer — golden-bad KEM ⇒ handshake fails), §11.4.50 (deterministic KEM-correctness over N runs), §11.4.150 (deep-research before the structural KEM pick), §11.4.93 (phases → workable items), §11.4.65/§11.4.153 (HTML+PDF+[DOCX] exports follow in refinement). Honest boundary (§11.4.6): the hybrid-composition formal proof rests on the WireGuard whitepaper, UNREACHABLE here — a human must verify it before relying on §3's IND-CCA argument.